THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Life Processes in Plants** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which mineral is needed by plants for making chlorophyll? 1

(a) sodium **(b)** silver **(c)** copper only **(d)** magnesium

2. The gas that collects as bubbles from hydrilla in sunlight is: 1

(a) oxygen **(b)** nitrogen **(c)** carbon dioxide **(d)** hydrogen

3. Which of the following reduces water loss in desert plants? 1

(a) large leaves **(b)** thick cuticle

(c) thin leaves **(d)** many stomata open

4. Which of the following is a food chain starting point? 1

(a) green plants **(b)** herbivores **(c)** carnivores **(d)** decomposers

5. In the starch test, the chlorophyll of the leaf is removed by boiling it in: 1

(a) alcohol **(b)** vinegar **(c)** milk **(d)** oil

6. A cactus makes food mostly in: 1

(a) the green stem **(b)** the spines **(c)** the flower **(d)** the roots

7. The ultimate source of energy for most food chains is: 1

(a) the Sun **(b)** the soil **(c)** the water **(d)** the air

8. Which statement about plant roots is correct? 1

(a) roots have stomata on tips

(b) roots make food

(c) roots absorb dissolved minerals and need oxygen

(d) roots release oxygen

9. **Assertion (A):** Pulse crops need less nitrogen fertiliser. 1

Reason (R): Rhizobium bacteria in their root nodules convert atmospheric nitrogen into usable form.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Cuscuta is a parasitic plant. 1
Reason (R): The pitcher plant traps insects to obtain nitrogen.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do farmers use green manure of legumes? 2

12. Why do potted plants near a window lean towards it? 2

13. Why is plant growth reduced when the leaves are covered with dust? 2

14. Why are stomata mostly present on the lower surface of most leaves? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the surface area of leaves and of root hairs is large. 3

16. Describe the role of guard cells in conserving water at night. 3

17. Explain the process by which a seed stores energy for germination. 3

OR

A plant is kept in the dark for 3 days and then for 3 days in light. Predict starch tests on a leaf after each period. 3

18. A student placed a plant in a bell jar with an oxygen-free atmosphere in the dark. Predict the effect on the plant. 3

OR

Discuss two examples where plants depend on other organisms for nutrition. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A farmer in Haryana notices a yellow, leafless creeper twining around a hedge. In a botanical garden he sees a pitcher plant, and in his pulse field he finds small swellings on the roots of the plants. 5
(i) What is the mode of nutrition of the yellow creeper?
(ii) Name the creeper.
(iii) Name the bacteria found in the root swellings.
(iv) Explain why the pitcher plant traps insects and why the hedge plant becomes weak.

20. Explain how a farmer can increase the yield of crops by using knowledge of photosynthesis, nitrogen fixation and transpiration. 5

OR

Design a plan for a school garden that uses plant processes (photosynthesis, transpiration, nitrogen fixation) to grow healthy vegetables. 5

END OF QUESTION PAPER — **Total: 40 Marks**